

THE ALGEBRAIC HODGE TO DE RHAM SPECTRAL SEQUENCE

YANGDA BEI

ABSTRACT. We provide an exposition of the proof degeneration of the algebraic Hodge to de Rham spectral sequence for a smooth and proper scheme X over a field k following [DI87]. The case in which k has prime characteristic is the primary focus of this paper, but we also provide a sketch for extending the proof to the general case.

1. INTRODUCTION

The study of algebraic geometry and complex analytic geometry are closely linked together. Similarities between the two fields have been known for a long time, but it was Serre who proved the correspondence between algebraic coherent sheaves and analytic coherent sheaves over a complete complex algebraic variety in [Ser56]. From this, we know that statements from complex geometry have an algebraic counterpart, but it is harder to translate the proofs of statements between the two.

The main result from Hodge theory is the Hodge decomposition theorem, which has an almost equivalent formulation as the degeneration of a spectral sequence, known as the Hodge to de Rham spectral sequence. This spectral sequence provides a relationship between the de Rham and Dolbeault cohomologies of a complex manifold. In particular, it degenerates on the first page for compact Kähler manifolds (see [Bei24] for an exposition by the author of its proof in the analytic case). Since compact Kähler manifolds are smooth complex projective varieties, one might wish to find a completely algebraic proof of the theorem in the algebraic case:

Theorem 1.1 ([DI87]). *If X is a smooth proper scheme over a field k , the spectral sequence induced by the Hodge filtration on the algebraic de Rham complex*

$$E_1^{p,q} = \mathbb{H}^{p,q}(X, \mathrm{gr}_p \mathrm{Fil}^\bullet \Omega_{X/k}^\bullet) \Rightarrow \mathbb{H}^{p+q}(X, \Omega_{X/k}^\bullet)$$

degenerates at the first page.

The aim of this paper is to present the proof of Theorem 1.1 following [DI87] and [Ill02], providing an exposition and filling in certain details left out by the original authors. In Section 2, we recall the definition of an algebraic differential, define and state some properties of morphisms, introduce sheaf cohomology and hypercohomology, and define Witt vectors. We then define the algebraic de Rham complex and the Hodge to de Rham spectral sequence in Section 3. Section 4 is an interlude on schemes of prime characteristic, and we prove the degeneration of the Hodge to de Rham spectral sequence in Section 5. We conclude the proof by generalising the argument to fields of characteristic zero in 6.

2. SOME PRELIMINARIES

We begin by recalling and defining several concepts that are needed to understand the statement of Theorem 1.1 and its proof.

Throughout the paper, if $X = \mathrm{Spec}(R)$ is affine, then we denote the affine n -space over X as $\mathbb{A}_X^n := \mathrm{Spec}(R[t_1, \dots, t_n])$. For general schemes X , let $\mathbb{A}_X^n := \mathbb{A}_{\mathbb{Z}}^n \times_{\mathrm{Spec}(\mathbb{Z})} X$.

Date: November 8, 2024.

2.1. Differentials. We begin this section by recalling basic definitions for differentials (or *Kähler* differentials).

Definition 2.1. Suppose A is a commutative ring, B is an A -algebra via the ring homomorphism $\varphi: A \rightarrow B$, and M is a B -module. An A -derivation on B is an A -module homomorphism $d: B \rightarrow M$ which satisfies the Leibniz rule

$$d(fg) = f dg + g df$$

for all $f, g \in B$. We denote the set of all such functions $\text{Der}_A(B, M)$.

Note that the condition $df = 0$ if $f \in A$ comes from the assumption that d is an A -module homomorphism.

Definition 2.2. Let $A \rightarrow B$ be a ring homomorphism. The module of *Kähler differentials* or the *module of relative differential forms* of B over A is the B -module $\Omega_{B/A}$ for which there is a *universal derivation* $d: B \rightarrow \Omega_{B/A}$ which satisfies the following universal property: for any B -module M and any derivation $d': B \rightarrow M$, there exists a unique B -module homomorphism $\varphi: \Omega_{B/A} \rightarrow M$ such that $d' = \varphi \circ d$.

Equivalently, let I be the kernel of the map $B \otimes_A B \rightarrow B$ defined by $\sum f_i \otimes g_i \mapsto \sum f_i \cdot g_i$. Then define

$$\Omega_{B/A} := I/I^2$$

and the universal derivation is the homomorphism d given by $df = 1 \otimes f - f \otimes 1$.

We thus have the isomorphism

$$\text{Hom}_B(\Omega_{B/A}, M) \cong \text{Der}_A(B, M)$$

We can carry this definition over to schemes. Let $X \rightarrow Y$ be a morphism of schemes, and let $\Delta: X \rightarrow X \times_Y X$ be the diagonal map. Then $\Delta(X)$ is locally closed subscheme of $X \times_Y X$, that is, a subscheme of an open subset W of $X \times_Y X$.

Definition 2.3. Let $\mathcal{I}$ be the sheaf of ideals of $\Delta(X)$ in W . We define the *sheaf of relative differentials* of X over Y to be the quasi-coherent pullback sheaf of $\mathcal{O}_X$ -modules $\Omega_{X/Y} = \Delta^*(\mathcal{I}/\mathcal{I}^2)$ on X .

2.2. Smooth, flat, and étale morphisms. Recall that a module M over a ring R is *flat* if the functor $-\otimes_R M$ is exact. A ring map $R \rightarrow A$ is said to be *flat* if A is flat as an R -module.

Definition 2.4 (Flat morphisms). Let $f: X \rightarrow Y$ be a morphism of schemes. We say that f is *flat* if for any point x of X , $\mathcal{O}_{X,x}$ is a flat module over $\mathcal{O}_{Y,f(x)}$.

We now define smooth and étale morphisms in the following way.

Definition 2.5. A morphism of schemes $i: T_0 \rightarrow T$ is a *thickening of order 1* if i is a closed immersion defined by an ideal of $\mathcal{O}_T$ of square zero.

Definition 2.6 (Smooth and étale morphisms). Let $f: X \rightarrow Y$ be a morphism of schemes. We say that f is *smooth* (resp. *étale*) if f is locally of finite presentation, and if for any commutative diagram

$$\begin{array}{ccc} & & X \\ & \nearrow g_0 & \downarrow f \\ T_0 & \xrightarrow{i} & T \longrightarrow Y \end{array}$$

where i is a thickening of order 1, there exists a (resp. a unique) Y -morphism $g: T \rightarrow X$ such that $g \circ i = g_0$ locally in the Zariski topology on T .

Lemma 2.7. Assume $f: X \rightarrow Y$ is a smooth map of schemes. Then f factors as

$$X \xrightarrow{g} \mathbb{A}_Y^n \xrightarrow{h} Y,$$

where h is the canonical projection and g is étale.

Proof. See [Ill02, 2.7] □

Proposition 2.8. *Let $f: X \rightarrow Y$, $g: Y \rightarrow S$ be morphisms of schemes. If f is smooth, then*

$$0 \longrightarrow f^* \Omega_{Y/S} \longrightarrow \Omega_{X/S} \longrightarrow \Omega_{X/Y} \longrightarrow 0$$

is exact. In particular, if f is étale, the canonical homomorphism $f^(\Omega_{Y/X}) \rightarrow \Omega_{X/S}$ is an isomorphism.*

Proof. See [Gro66a, 17.2.3] or [Ill02, 2.5]. □

The second assertion in Proposition 2.8 has a converse, which is deduced in a similar manner.

Proposition 2.9. *In the same situation as 2.8 (b), assume gf is a smooth map. If the canonical homomorphism $f^* \Omega_{Y/S} \rightarrow \Omega_{X/S}$ is an isomorphism, then f is étale.*

Definition 2.10. Let $f: X \rightarrow Y$ be a morphism of schemes. A *geometric fibre* of f is the reduced scheme of a fibre $X_y = X \times_Y \text{Spec } k(y)$ of f at a point y by an extension of scalars to an algebraic closure of $k(y)$.

Proposition 2.11. *Let $f: X \rightarrow Y$ be a morphism of schemes, locally of finite presentation. The following conditions are equivalent:*

- (i) f is smooth
- (ii) f is flat and the geometric fibres of f are regular schemes.

Proof. This is a verification of the discussion found in 2.7 and 2.8 of [Ill02]. □

Definition 2.12. If $f: X \rightarrow Y$ is smooth, then the integer

$$\dim_x(f) := \dim_{k(x)} \Omega_{X/Y}^1 \otimes k(x)$$

is called the relative dimension of f at x . Furthermore, f is of *pure relative dimension* r if the relative dimension of f is constant equal to the integer r .

2.3. Sheaf cohomology and hypercohomology. We now define the sheaf cohomology of a sheaf $\mathcal{F}$ over a topological space X . We shall see that every sheaf has a canonical resolution, which allows us to obtain a well-defined sheaf cohomology theory. However, standard sheaf cohomology may not capture all the necessary information in some settings, so we will see that hypercohomology is a natural generalisation of sheaf cohomology, allowing us to address these gaps.

Fix an open subset U of X . For every sheaf $\mathcal{F}$ on X , we can associate the abelian group $\Gamma(U, \mathcal{F}) := \mathcal{F}(U)$ and to every sheaf morphism $\phi: \mathcal{F} \rightarrow \mathcal{G}$ the group homomorphism $\phi_U: \mathcal{F}(U) \rightarrow \mathcal{G}(U)$. This makes $\Gamma(U, -)$ into a functor from the category of sheaves to abelian groups, henceforth called the *sections functor*. Aptly, we call $\Gamma(X, -)$ the *global sections functor*.

Lemma 2.13. *The global sections functor is left-exact.*

Let $\mathcal{C}^0 \mathcal{F}$ be the sheaf defined by $\mathcal{C}^0 \mathcal{F}(U) := \bigoplus_{x \in U} \mathcal{F}_x$. If $\mathcal{Q}^1$ is the quotient sheaf of $\mathcal{C}^0 \mathcal{F}$ of $\mathcal{F}$, then we have a short exact sequence of sheaves

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{C}^0 \mathcal{F} \longrightarrow \mathcal{Q}^1 \longrightarrow 0.$$

Then let $\mathcal{C}^1 \mathcal{F} := \mathcal{C}^0 \mathcal{Q}^1$ and $\mathcal{Q}^2$ be the quotient of $\mathcal{C}^0 \mathcal{Q}^1$ by $\mathcal{Q}^1$, which yields the short exact sequence

$$0 \longrightarrow \mathcal{Q}^1 \longrightarrow \mathcal{C}^1 \mathcal{F} \longrightarrow \mathcal{Q}^2 \longrightarrow 0.$$

We can join the two short exact sequences into a four-term exact sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{C}^0 \mathcal{F} & \dashrightarrow & \mathcal{C}^1 \mathcal{F} \longrightarrow \mathcal{Q}^2 \longrightarrow 0. \\ & & & & \searrow & & \nearrow \\ & & & & & \mathcal{Q}^1 & \end{array}$$

Repeating the construction on $\mathcal{Q}^2$, and so on, we obtain the long exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{C}^0 \mathcal{F} \xrightarrow{\delta} \mathcal{C}^1 \mathcal{F} \xrightarrow{\delta} \mathcal{C}^2 \mathcal{F} \xrightarrow{\delta} \dots$$

of sheaves. This yields the *Godement canonical resolution* of $\mathcal{F}$, and we call $\mathcal{C}^i(-)$ the *i-th Godement functor*. We can now state the definition of sheaf cohomology.

Definition 2.14. The *sheaf cohomology groups* $H^i(X, \mathcal{F})$ of a sheaf of abelian groups $\mathcal{F}$ on a topological space is

$$H^i(X, \mathcal{F}) = \frac{\ker \delta: \mathcal{C}^i \mathcal{F}(X) \rightarrow \mathcal{C}^{i+1} \mathcal{F}(X)}{\operatorname{im} \delta: \mathcal{C}^{i-1} \mathcal{F}(X) \rightarrow \mathcal{C}^i \mathcal{F}(X)}.$$

This construction gives a concrete definition of sheaf cohomology. We can also wish to define it as follows:

Definition 2.15. The *sheaf cohomology groups* $H^i(X, \mathcal{F})$ of a sheaf of abelian groups $\mathcal{F}$ on a topological space is

$$H^i(X, \mathcal{F}) := R^i \Gamma(X, \mathcal{F}),$$

where $R^i \Gamma$ is the *i-th* (right-)derived functor of Γ .

Lemma 2.16. *Different choices of injective resolutions give an isomorphism between the objects $R^i \Gamma(\mathcal{F})$.*

Proof. See [Voi02, Proposition 4.30]. □

The hypercohomology of a complex of sheaves $\mathcal{F}^\bullet$ generalises the cohomology of a single sheaf. As with normal sheaf cohomology, we can compute the hypercohomology of a complex of sheaves by taking the Godement canonical resolution, this time for every sheaf $\mathcal{F}^\bullet$. This yields the double complex

$$K = \bigoplus_{p,q} K^{p,q} = \bigoplus_{p,q} \Gamma(X, \mathcal{C}^p \mathcal{F}^q)$$

with the horizontal differential $\delta: K^{p,q} \rightarrow K^{p+1,q}$ induced from the Godement resolution and the vertical differential $d: K^{p,q} \rightarrow K^{p,q+1}$ induced from the complex $\mathcal{F}^\bullet$. The fact that these two differentials commute follows since the differential $d: \mathcal{F}^q \rightarrow \mathcal{F}^{q+1}$ induces a morphism of complexes $\mathcal{C}^\bullet \mathcal{F}^1 \rightarrow \mathcal{C}^\bullet \mathcal{F}^{q+1}$ and that the Godement functor $\mathcal{C}^\bullet(-)$ from sheaves to sheaves is exact. We can define hypercohomology as follows:

Definition 2.17. The *hypercohomology* $\mathbb{H}^i(X, \mathcal{F}^\bullet)$ of a complex of sheaves $\mathcal{F}^\bullet$ over a topological space X is the cohomology of the double complex K , that is the single cohomology $H_D^i(K^\bullet)$ of the complex

$$K^\bullet = \bigoplus_i K^i = \bigoplus_i \bigoplus_{p+q=i} K^{p,q}$$

with differential $D = \delta + (-1)^p d$ as defined above.

If $\mathcal{F}^\bullet$ is the complex with a single sheaf $\mathcal{F}_0$ in degree 0 given by

$$0 \longrightarrow \mathcal{F}^0 \longrightarrow 0 \longrightarrow 0 \longrightarrow \dots,$$

then $\bigoplus K^{p,q} = \bigoplus \Gamma(X, \mathcal{C}^p \mathcal{F}^q)$ has non-zero entries in only the zeroth row. This is simply the complex of sections of the Godement resolution of $\mathcal{F}^0$, so in this case, $\mathbb{H}^i(X, \mathcal{F}^\bullet) = H^i(X, \mathcal{F}^0)$. It is in this sense that hypercohomology is a generalisation of sheaf cohomology.

We can similarly define the hypercohomology of X with coefficients in $\mathcal{F}^\bullet$ in terms of derived functors:

Definition 2.18. The *hypercohomology* $\mathbb{H}^i(X, \mathcal{F}^\bullet)$ of a complex of sheaves $\mathcal{F}^\bullet$ over a topological space X is

$$\mathbb{H}^i(X, \mathcal{F}^\bullet) := \mathbb{R}^i \Gamma(X, \mathcal{I}^\bullet),$$

where $\mathcal{I}^\bullet$ is an injective resolution of $\mathcal{F}^\bullet$.

Again, the double complexes of sheaves with different injective resolutions are quasi-isomorphic (see [Voi02, Corollary 8.9]).

2.4. Witt vectors. The theory of Witt vectors arises from the study of properties of certain p -adic rings. For our purposes, the following definition and example suffices.

Definition 2.19. Suppose k is a perfect field of characteristic p . The *length two Witt vectors of k* , denoted by $W_2(k)$, is the ring k^2 with addition and multiplication defined as

$$\begin{aligned} (a_1, a_2) + (b_1, b_2) &= (a_1 + b_1, a_2 + b_2 + p^{-1}(a_1^{p-1} + b_1^{p-1} - (a_1 + b_1)^p)) \\ (a_1, a_2)(b_1, b_2) &= (a_1 b_1, b_1^p a_2 + b_2 a_1^p) \end{aligned}$$

Example 2.20. If $k = \mathbb{F}_p$, then $W_2(k) \cong \mathbb{Z}/p^2\mathbb{Z}$ given by the isomorphism $(a_1, a_2) \mapsto \tau(a_1)p\tau(a_2)$, where τ denotes the multiplicative section of $\mathbb{Z}/p^2\mathbb{Z} \rightarrow \mathbb{F}_p$.

3. COHOMOLOGY AND THE HODGE TO DE RHAM SPECTRAL SEQUENCE

3.1. Algebraic de Rham cohomology. Like its counterparts on smooth manifolds, we can define the exterior powers of differentials in a similar manner. A *differential p -form* are elements of the p -th exterior power over $\mathcal{O}_X$

$$\Omega_{X/Y}^p := \bigwedge^p \Omega_{X/Y}^1,$$

that is, $\Omega_{X/Y}^p$ is the free $\mathcal{O}_X$ -module on symbols $\omega_1 \wedge \cdots \wedge \omega_p$ with $\omega_i \in \Omega_{X/Y}^1$, modulo relations of the form

$$(f\omega_1 + f'\omega'_1) \wedge \cdots \wedge \omega_p - r(\omega_1 \wedge \cdots \wedge \omega_p) - f'(\omega'_1 \wedge \cdots \wedge \omega_p)$$

(and similarly for each position) and $\omega_1 \wedge \cdots \wedge \omega_p$ whenever two of the ω_i are equal. The map $d: \mathcal{O}_X \rightarrow \Omega_{X/Y}^1$ induces a map $d: \Omega_{X/Y}^p \rightarrow \Omega_{X/Y}^{p+1}$ by

$$f\omega_1 \wedge \cdots \wedge \omega_p \mapsto df \wedge \omega_1 \wedge \cdots \wedge \omega_p.$$

One can check that $d \circ d = 0$, so we get the cochain complex

$$0 \longrightarrow \mathcal{O}_X \xrightarrow{d} \Omega_{X/Y}^1 \xrightarrow{d} \Omega_{X/Y}^2 \xrightarrow{d} \cdots$$

known as the *algebraic de Rham complex*. We denote $\Omega_{X/Y}^0 := \mathcal{O}_X$. One might be tempted to take the cohomology of this complex as the definition of the algebraic de Rham cohomology. However, we see in the following example that this definition is too naive.

Example 3.1. Let X be a complex manifold and Ω_{hol}^p be the sheaf of holomorphic p -forms on X . The holomorphic forms of X make up the *holomorphic de Rham complex* with differential $\partial: \Omega_{\text{hol}}^p \rightarrow \Omega_{\text{hol}}^{p+1}$. Then the isomorphism

$$H_{\text{sing}}^i(X; \mathbb{C}) \cong h^i(\Omega_{\text{hol}}^\bullet(X))$$

cannot hope to hold, where h^i is the i -th cohomology group of the complex of global sections complex $\Omega_{\text{hol}}^\bullet(X)$. In particular, for $i \geq \dim_{\mathbb{C}}(X)$, we see that $H^i(X, \Omega_{X/\mathbb{C}}^k) = 0$ but $H_{\text{sing}}^i(X; \mathbb{C}) = 0$ for only $i > \dim_{\mathbb{R}}(X) = 2 \cdot \dim_{\mathbb{C}}(X)$, so the complex $\Omega^\bullet$ is too short.

We thus give the following definition.

Definition 3.2. Suppose X is a finite-dimensional scheme over a field k . The *algebraic de Rham cohomology* of X is the hypercohomology of the algebraic de Rham complex

$$H_{\text{dR}}^i(X/k) := \mathbb{H}^i(X, \Omega_{X/k}^\bullet).$$

To see that this is indeed the “correct” definition, we hope to recover the isomorphism

$$H_{\text{sing}}^i(X; \mathbb{C}) \cong H_{\text{dR}}^i(X/\mathbb{C}).$$

Given some sheaf $\mathcal{F}$ and a resolution $\mathcal{F}^\bullet$ of $\mathcal{F}$

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}^0 \longrightarrow \mathcal{F}^1 \longrightarrow \dots,$$

that is an exact sequence $\mathcal{F}^\bullet$ such that $\mathcal{F} \rightarrow \mathcal{F}^0$ is injective, it is easy to see that the complexes

$$0 \longrightarrow \mathcal{F} \longrightarrow 0 \longrightarrow 0 \longrightarrow \dots \quad \text{and} \quad 0 \longrightarrow \mathcal{F}^0 \longrightarrow \mathcal{F}^1 \longrightarrow \mathcal{F}^2 \longrightarrow \dots$$

are quasi-isomorphic, which gives us that

$$H^i(X, \mathcal{F}) \cong \mathbb{H}^i(X, \mathcal{F}) \cong \mathbb{H}^i(X, \mathcal{F}^\bullet).$$

For a smooth variety X over $\mathbb{C}$, we then have

$$H_{\text{sing}}^i(X; \mathbb{C}) \cong H^i(X, \underline{\mathbb{C}}) \cong \mathbb{H}^i(X, \Omega_{X/\mathbb{C}}^\bullet) = H_{\text{dR}}^i(X/\mathbb{C}),$$

as desired, where $\underline{\mathbb{C}}$ denotes the constant sheaf on X whose stalks are all $\mathbb{C}$. The first isomorphism is due to [Gro66a].

3.2. Hodge cohomology. In this subsection, we define the Hodge cohomology groups.

Definition 3.3. Let X be a scheme, k a field, and $f: X \rightarrow \text{Spec}(k)$ a morphism of schemes. The *Hodge cohomology* of a variety X/k is

$$H_{\text{Hodge}}^\ell(X/k) := \bigoplus_{i+j=\ell} H^i(X, \Omega_{X/k}^j),$$

where $H^i(X, \Omega_{X/k}^j)$ is the i -th sheaf cohomology of $\Omega_{X/k}^j$ on X .

Proposition 3.4. Any quasi-isomorphism of complexes $\alpha: \mathcal{F}^\bullet \rightarrow \mathcal{G}^\bullet$ induces an isomorphism $\alpha_*: \mathbb{H}^i(X, \mathcal{F}^\bullet) \rightarrow \mathbb{H}^i(X, \mathcal{G}^\bullet)$.

Proposition 3.5. Let $\mathcal{F}$ denote a complex in which every term is 0 except for at $i = 0$ where it is $\mathcal{F}$. Then $\mathbb{H}^i(X, \mathcal{F}) = \mathbb{H}^i(X, \mathcal{F})$.

3.3. The Hodge to de Rham spectral sequence. Let X/k be a smooth proper scheme. Consider the algebraic de Rham complex equipped with the filtration

$$\text{Fil}^p \Omega^\bullet = (\dots \longrightarrow 0 \longrightarrow \Omega_{X/k}^p \longrightarrow \Omega_{X/k}^{p+1} \longrightarrow \Omega_{X/k}^{p+2} \longrightarrow \dots).$$

This naive, or “stupid”, filtration is called the *Hodge filtration*. This induces the spectral sequence given by

$$E_1^{p,q} = \mathbb{H}^{p,q}(X, \text{gr}_p \text{Fil}^\bullet \Omega_{X/k}^\bullet) \Rightarrow \mathbb{H}^{p+q}(X, \Omega_{X/k}^\bullet).$$

In our particular case, it becomes the *Hodge*, or *algebraic Frölicher*, spectral sequence:

$$E_1^{p,q} = H^q(X, \Omega_{X/k}^p) \Rightarrow H_{\text{dR}}^{p+q}(X/k).$$

The following lemma will be crucial in proving the degeneration of the Hodge to de Rham spectral sequence.

Lemma 3.6. Let X be a proper scheme over a field k . Then $H_{\text{Hodge}}^\ell(X/k) \cong H_{\text{dR}}^\ell(X/k)$ if and only if

$$\sum_{i+j=\ell} \dim_k H^j(X, \Omega_{X/k}^i) = \dim_k H_{\text{dR}}^\ell(X/k).$$

Proof. Since X is proper, $H_{\text{Hodge}}^\ell(X/k)$ and $H_{\text{dR}}^\ell(X/k)$ are finite-dimensional (see [Har13, III 5.2] for the projective case, [Gro61, III 3] for the general case). This means that $H_{\text{Hodge}}^\ell(X/k) \cong H_{\text{dR}}^\ell(X/k)$ if and only if

$$\dim_k H_{\text{dR}}^\ell(X/k) = \dim_k H_{\text{Hodge}}^\ell(X/k) = \sum_{i+j=\ell} \dim_k H^j(X, \Omega_{X/k}^i). \quad \square$$

4. SCHEMES OF PRIME CHARACTERISTIC

One typically proves the equivalence of Hodge and algebraic de Rham cohomology for schemes by first proving it for complex manifolds and then extending it to schemes over fields of characteristic zero via Serre's GAGA principle. However, this method fails when k is an arbitrary field. We now introduce the Frobenius map and Cartier isomorphism, tools which will be used to prove the degeneration of the algebraic Hodge to de Rham spectral sequence.

Throughout this section, let p denote a fixed prime number.

4.1. Frobenius morphisms. We say a scheme X is of *prime characteristic* p if $p\mathcal{O}_X = 0$, that is, if the morphism $X \rightarrow \text{Spec}(\mathbb{Z})$ factors uniquely through $\text{Spec}(\mathbb{F}_p)$. This means that for all open $U \subset X$, we have that $p\mathcal{O}_X(U)$ is the zero ring, making $\mathcal{O}_X(U)$ a ring of characteristic p .

Definition 4.1. If X is a scheme of prime characteristic, then the *absolute Frobenius morphism* of X (or simply *Frobenius endomorphism*) to be the endomorphism $F_X: X \rightarrow X$ that is the identity on the underlying space $|X|$, and the p -th power morphism on structure sheaves. On $U = \text{Spec}(R) \subset X$, F_X corresponds to the Frobenius endomorphism $a \mapsto a^p$ on R .

If $f: X \rightarrow Y$ is a morphism of schemes, then there is a commutative square

$$(4.1) \quad \begin{array}{ccc} X & \xrightarrow{F_X} & X \\ \downarrow f & & \downarrow f \\ Y & \xrightarrow{F_Y} & Y \end{array}$$

simply because if $\varphi: A \rightarrow B$ is a homomorphism of rings of characteristic p , then $\varphi(a^p) = \varphi(a)^p$. This allows us to construct the fibre-product $X^{(p)} := Y \times_{(Y, F_Y)} X$ induced from X by the base change F_Y . We call $X^{(p)}$ the *Frobenius twist* of X . The morphism F_X defines a unique Y -morphism $F_{X/Y}: X \rightarrow X^{(p)}$, called the *relative Frobenius morphism* of X , which gives rise to the commutative diagram

$$(4.2) \quad \begin{array}{ccccc} X & & \xrightarrow{F_X} & & X \\ & \searrow F_{X/Y} & & & \downarrow f \\ & & X^{(p)} & \xrightarrow{\quad} & X \\ & \downarrow f & \downarrow & & \downarrow f \\ & & Y & \xrightarrow{F_Y} & Y \end{array}$$

Concretely, if $Y = \text{Spec}(R)$ is affine, and $X = \mathbb{A}_Y^n$, then morphisms $X \rightarrow X^{(p)}$ and $X^{(p)} \rightarrow X$ correspond respectively to the homomorphisms $t_i \mapsto t_i^p$ and $at_i \mapsto a^p t_i$ for $a \in R$. If X/Y is smooth, then so is $X^{(p)}$ being the base change along the absolute Frobenius map of Y . We also note that if Y is a perfect scheme, that is if F_Y is an automorphism, for example if $Y = \text{Spec}(k)$ where k is a perfect field of characteristic p , then $X^{(p)} \rightarrow X$ is an isomorphism of schemes but not of schemes over Y .

From here, we denote $F := F_{X/Y}$ where no confusion arises. As it turns out, the Frobenius twist has no effect on cohomology.

Lemma 4.2. *Let k be a perfect field of characteristic p and $X/\mathrm{Spec}(k)$ be smooth projective. Then*

$$H_{\mathrm{Hodge}}^\ell(X/k) \cong H_{\mathrm{Hodge}}^\ell(X^{(p)}/k).$$

Proof of Lemma 4.2. We have morphisms $f: X^{(p)} \rightarrow X$, $g: X \rightarrow \mathrm{Spec}(k)$ where f is an isomorphism, and hence smooth. This gives us an exact sequence

$$0 \longrightarrow f^*\Omega_{X/k} \longrightarrow \Omega_{X/k} \longrightarrow \Omega_{X^{(p)}/X} \longrightarrow 0$$

where $\Omega_{X^{(p)}/X} = 0$ since f is an isomorphism. This means that $f^*\Omega_{X/k} \cong \Omega_{X/k}$, so by taking wedge products, we obtain isomorphisms $f^*\Omega_{X/k}^i \cong \Omega_{X/k}^i$ of $\mathcal{O}_{X^{(p)}}$ -modules for each $i \geq 0$. This gives us an isomorphism on sheaf cohomology

$$H^j(X, \Omega_{X/k}^i) \cong H^j(X^{(p)}, f^*\Omega_{X/k}^i) \cong H^j(X^{(p)}, \Omega_{X^{(p)}/k}^i),$$

where the second isomorphism follows from the fact that f is an isomorphism. Taking direct sums gives the desired result. $\square$

Proposition 4.3. *Let Y be a scheme of characteristic p and $f: X \rightarrow Y$ of pure relative dimension n . Then the relative Frobenius morphism $F: X \rightarrow X^{(p)}$ is a finite and flat morphism, and the $\mathcal{O}_{X^{(p)}}$ -module $F_*\mathcal{O}_X$ is locally free of rank p^n . In particular, if f is étale, F is an isomorphism.*

Proof. Suppose first that $n = 0$. Since F is a smooth map of relative dimension 0, it is étale [Gro66a, 17.10.2], which means it is flat and finite [Gro66a, 17.6.2]. The case in which $X = \mathbb{A}_Y^n = \mathrm{Spec}(Y[t_1, \dots, t_n])$ follows since the monomials $\prod t_i^{m_i}$ with $0 \leq m_i < p-1$ form a basis for $F_*\mathcal{O}_X$ over $\mathcal{O}_X$. In general, we can factor f as maps $g: X \rightarrow \mathbb{A}_Y^n$ and $h: \mathbb{A}_Y^n \rightarrow Y$ according to Lemma 2.7, where h is the canonical projection map, which is smooth and of pure relative dimension n . The map g is étale by construction, and therefore finite and flat. $\square$

We see that as a result of the following lemma, the differential of the complex $F_*\Omega_{X/Y}^\bullet$ is $\mathcal{O}_{X^{(p)}}$ -linear.

Lemma 4.4. *Let Y be a scheme of characteristic p and $f: X \rightarrow Y$ a morphism. The canonical homomorphisms associated to the relative and absolute Frobenius maps $F_X^*\Omega_{X/Y}^1 \rightarrow \Omega_{X/Y}^1$ and $F^*\Omega_{X/Y}^1 \rightarrow \Omega_{X/Y}^1$ are zero.*

Proof. If s is a local section of $\mathcal{O}_X$, it follows that $d(s^p) = ps^{p-1}ds = 0$. Since $d(s^p) = F_X^*(ds) = F^*(1 \otimes ds)$, the result follows. $\square$

4.2. Cartier isomorphism. We now state and sketch the proof of to Cartier isomorphism.

Theorem 4.5 (Cartier). *Let Y be a scheme of characteristic p and $f: X \rightarrow Y$ a morphism.*

(a) *There exists a unique homomorphism of graded $\mathcal{O}_X$ -algebras*

$$C^{-1}: \bigoplus \Omega_{X^{(p)}/Y}^i \rightarrow \bigoplus \mathcal{H}^i F_*\Omega_{X/Y}^\bullet$$

satisfying the following two conditions:

(i) *for $i = 0$, γ is given by the homomorphism $F^*: \mathcal{O}_{X^{(p)}} \rightarrow F_*\mathcal{O}_X$;*

(ii) *for $i = 1$, γ sends $1 \otimes ds$ to the class of $s^{p-1}ds$ in $\mathcal{H}^1 F_*\Omega_{X/Y}^\bullet$ (where $1 \otimes ds$ denotes the image of the section ds of $\Omega_{X/Y}$ in $\Omega_{X^{(p)}/Y}$).*

(b) *If f is smooth, γ is an isomorphism.*

Here, $\mathcal{H}^i F_*\Omega_{X/k}^\bullet$ denotes the cohomology of the complex $F_*\Omega_{X/Y}^\bullet$ of $\mathcal{O}_{X^{(p)}}$ -modules. We call C^{-1} the *Cartier isomorphism* in the case of (b).

Proof. We briefly outline a sketch for the proof to (a) following [Kat70]. To show that there exists a unique homomorphism γ , it suffices to show that there exists a homomorphism of graded $\mathcal{O}_X$ -algebras

$$C_{\text{abs}}^{-1}: \bigoplus \Omega_{X/Y}^i \rightarrow \bigoplus \mathcal{H}^i F_{X*} \Omega_{X/Y}^\bullet$$

such that γ_{abs} is given by F_X^* for $i = 0$, γ_{abs} is given by sending ds in $\Omega_{X/Y}$ to $s^{p-1}ds$ for $i = 1$. We call C^{-1}_{abs} the *absolute Cartier morphism*. This is justified since any commutative square

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

defines a canonical homomorphism of $\mathcal{O}_{X'}$ -modules $g^* \Omega_{X/Y} \rightarrow \Omega_{X'/Y'}$, which sends $1 \otimes g^{-1}(d_{X/Y}s)$ to $d_{X'/Y'}(1 \otimes g^{-1}(s))$. However, the map of $\mathcal{O}_X$ in $\mathcal{H}^1(F_{X*} \Omega_{X/Y}^\bullet)$ sending a local section s of $\mathcal{O}_X$ onto the class of $s^{p-1}ds$ is a Y -derivation, which follows as a result of the identity

$$\frac{1}{p}((X+Y)^p - X^p - Y^p) = \frac{1}{p} \sum_{0 \leq i \leq p} \binom{p}{i} X^{p-i} Y^i$$

in $\mathbb{Z}[X, Y]$. We thus have the desired homomorphism for γ_{abs} in degree 0. The exterior algebra $\bigoplus \Omega_{X/Y}^i$ is not only anti-commutative but elements of odd degree are of square zero, and hence likewise of its sub-quotient $\bigoplus \mathcal{H}^i F_{X*} \Omega_{X/Y}^\bullet$. Consequently, there exists a unique homomorphism of graded algebras satisfying (i) and (ii).

For (b), we can factor f through $\mathbb{A}_Y^n$ as $X \xrightarrow{g} \mathbb{A}_Y^n \xrightarrow{h} Y$ by Lemma 2.7, where h is the canonical projection and g is étale. From 4.1, we get the commutative square

$$(4.3) \quad \begin{array}{ccc} X & \xrightarrow{F_X} & X^{(p)} \\ \downarrow g & & \downarrow g^{(p)} \\ Z & \xrightarrow{F_Y} & Z^{(p)}, \end{array}$$

where $Z = \mathbb{A}_Y^n$. By Proposition 2.8, $g^* \Omega_{Z/X} \rightarrow \Omega_{X/Y}$ is an isomorphism. The diagram above being a pullback square and the fact that the relative Frobenius map F is finite by Proposition 4.3, the map of complexes of $\mathcal{O}_X$ -modules induced by $g^{(p)}$

$$g^{(p)*} F_* \Omega_{Z/Y}^\bullet \rightarrow F_* \Omega_{X/Y}^\bullet$$

is an isomorphism. Since $g^{(p)}$ is étale, and therefore flat, the induced map on cohomology

$$g^{(p)*} \mathcal{H}^i F_* \Omega_{Z/Y}^\bullet \rightarrow \mathcal{H}^i F_* \Omega_{X/Y}^\bullet$$

is an isomorphism. On the other hand, since $g^{(p)*} \Omega_{Z'/Y}^i \rightarrow \Omega_{X'/Y}^i$ is an isomorphism as $g^{(p)}$ is étale, it follows that it suffices to prove (b) for Z by functoriality of γ . By analogous arguments using extension of scalars and Künneth's formula, we can reduce $Y = \text{Spec}(\mathbb{F}_p)$ and $n = 1$, that is, $Z = \text{Spec}(\mathbb{F}_p[t])$ (see [DI87, 3.7] for more details). Then $Z^{(p)} = Z$ and the monomials $1, t, \dots, t^{p-1}$ form a basis of $F_* \mathcal{O}_Z$ over $\mathcal{O}_Z$. Since the differential $d: F_* \mathcal{O}_Z \rightarrow F_* \mathcal{O} = (F_* \mathcal{O}_Z)dt$ sends t^i to it^{i-1} , one concludes that $\mathcal{H}^0 F_* \Omega_{Z/\mathbb{F}_p}^\bullet$ (resp. $\mathcal{H}^1 F_* \Omega_{Z/\mathbb{F}_p}^\bullet$) is free over $\mathcal{O}_Z$ with basis 1 (resp. $t^{p-1}dt$), and therefore, that γ is an isomorphism. $\square$

5. DEGENERATION OF THE HODGE TO DE RHAM SPECTRAL SEQUENCE

As a corollary to Theorem 4.5, the complexes $(\Omega_{X^{(p)}/k}^\bullet, 0)$ and $(F_*\Omega_{X/k}^\bullet, d)$ have isomorphic cohomology. Thus, it is natural to ask whether there exists a quasi-isomorphism $(\Omega_{X^{(p)}/k}^\bullet, 0) \rightarrow (F_*\Omega_{X/k}^\bullet, d)$ that induces the Cartier isomorphism on cohomology. This is a key step in proving the degeneration of the Hodge to de Rham spectral sequence.

Proposition 5.1. *Let X be a scheme such that there exists a quasi-isomorphism $(\Omega_{X^{(p)}/k}^\bullet, 0) \rightarrow (F_*\Omega_{X/k}^\bullet, d)$. Then the Hodge to de Rham spectral sequence degenerates on the first page.*

Proof. If there exists such a quasi-isomorphism, then

$$\begin{aligned} H_{\text{Hodge}}^\ell(X/k) &\stackrel{(1)}{\cong} H_{\text{Hodge}}^\ell(X^{(p)}/k) \cong \mathbb{H}^\ell(X^{(p)}, \Omega_{X^{(p)}/k}^\bullet) \\ &\stackrel{(2)}{\cong} \mathbb{H}^\ell(X^{(p)}, F_*\Omega_{X/k}^\bullet) \stackrel{(3)}{\cong} \mathbb{H}^\ell(X, \Omega_{X/k}^\bullet) \\ &\cong H_{\text{dR}}^\ell(X/k), \end{aligned}$$

whereby (1) follows from Lemma 4.2, (2) follows from the quasi-isomorphism assumption, and (3) follows from the fact that F is a finite map, so it has no higher pushforward. $\square$

Theorem 5.2. *Let S be a scheme of characteristic p and X a smooth and proper S -scheme. If $\tilde{S}$ is a scheme over $\mathbb{Z}/p^2\mathbb{Z}$ such that $X^{(p)}$ admits a smooth proper lift $\tilde{X}^{(p)}$ over $\tilde{S}$, then there is a quasi-isomorphism*

$$\tau_{<p}(\Omega_{X^{(p)}/S}^\bullet, 0) \rightarrow \tau_{<p}(F_*\Omega_{X/S}^\bullet, d)$$

inducing the Cartier isomorphism on cohomology, where the truncation $\tau_{<p}$ replaces the complex with 0 in positions $i \geq p$ and the kernel of the $(p-1)$ -th element in position $(p-1)$.

Proof. We first note that $\tau_{<p}(\Omega_{X^{(p)}/S}^\bullet, 0)$ with the zero differential is quasi-isomorphic to $\bigoplus_{i < p} \Omega_{X^{(p)}/S}^i[-i]$ as a consequence of the discussion after Definition 3.2. The aim is to construct morphisms

$$\psi^i = \psi_{\tilde{F}}^i: \Omega_{X^{(p)}/S}^i[-i] \rightarrow F_*\Omega_{X/S}^\bullet$$

as maps of complexes. In degree 0, we simply set $\psi^0 := F^*: \mathcal{O}_X \rightarrow F_*\mathcal{O}_X$. We proceed by first considering a global lift of X in constructing ψ^1 (step 1), then showing that different lifts induce a chain-homotopy on the complexes, and thus a quasi-isomorphism between them (step 2), generalising to an arbitrary scheme where we need to consider lifts on affine pieces (step 3), and finally extending ψ^1 to all ψ^i (step 4).

Step 1. Let X lift to $\tilde{X}$ and the Frobenius map F lift to $\tilde{F}: \tilde{X} \rightarrow \tilde{X}^{(p)}$ over $\tilde{S}$. By Lemma 4.4, $\tilde{F}$ induces the zero map $\tilde{F}^*: \Omega_{\tilde{X}^{(p)}/\tilde{S}}^1 \rightarrow \tilde{F}_*\Omega_{\tilde{X}/\tilde{S}}^1$. This yields the commutative diagram

$$(5.1) \quad \begin{array}{ccc} \Omega_{\tilde{X}^{(p)}/\tilde{S}}^1 & \xrightarrow{\tilde{F}^*} & p\tilde{F}_*\Omega_{\tilde{X}/\tilde{S}}^1 \\ \downarrow & & \uparrow p \\ \Omega_{X^{(p)}/S}^1 & \xrightarrow{\psi^1} & F_*\Omega_{X/S}^1, \end{array}$$

where the left vertical map is reduction modulo p . It is clear to see that $\tilde{F}^*$ factors through ψ^1 . Now, for any $dx \in \Omega_{X^{(p)}/S}^1$ where x is a local section of $\mathcal{O}_{X^{(p)}}$, let $\tilde{x}$ be a lift of x to $\mathcal{O}_{\tilde{X}^{(p)}}$, so $\tilde{F}(\tilde{x}) = \tilde{x}^p + p\tilde{y}$ for some $\tilde{y} \in \mathcal{O}_{\tilde{X}^{(p)}}$. As $\tilde{F}^*(d\tilde{X}) = p\tilde{x}^{p-1}d\tilde{x} + p d\tilde{y}$, we have that $\psi^1(dx) = x^{p-1}dx + dy$, and clearly $d(\psi^1 dx) = 0$. It is also clear that $\psi^1 d(dx) = 0$, so ψ^1 is a chain map.

Step 2. Next, suppose that there are different lifts $\tilde{F}_i: \tilde{X} \rightarrow \tilde{X}^{(p)}$ of the relative Frobenius map F , which gives rise to the maps $\psi_i^1: \Omega_{X^{(p)}/S}^1 \rightarrow F_*\Omega^1 X/S$. We claim that there exists a chain homotopy $H_{ij}: \Omega_{X^{(p)}/S}^1 \rightarrow F_*\Omega_{X/S}^0 = F_*\mathcal{O}_X$ for each pair $\tilde{F}_i$ and $\tilde{F}_j$, which satisfies $H_{ij} + H_{jk} = H_{ik}$. Observe that the map

$$\tilde{F}_i^* - \tilde{F}_j^*: \mathcal{O}_{\tilde{X}^{(p)}} \rightarrow \tilde{F}_{i*}\mathcal{O}_{\tilde{X}} \cong \tilde{F}_{j*}\mathcal{O}_{\tilde{X}}$$

has image in $p\tilde{F}_{i*}\mathcal{O}_{\tilde{X}} \cong p\tilde{F}_{j*}\mathcal{O}_{\tilde{X}} \xrightarrow{\sim} F_*\mathcal{O}_X$, thus inducing a map $\eta_{ij}: \mathcal{O}_{X^{(p)}} \rightarrow F_*\mathcal{O}_X$. It can be verified that η is an $\mathcal{O}_S$ -linear derivation, and hence induces the chain homotopy $H_{ij}: \Omega_{X^{(p)}/S}^1 \rightarrow F_*\mathcal{O}_X$, that is, $\psi_i^1 - \psi_j^1 = dH_{ij}$. It is also easy to verify that $H_{ij} + H_{jk} = H_{ik}$.

Step 3. Now let X be an arbitrary scheme covered by a set $\mathcal{U}$ of open affines, so that the Frobenius map on each $U_{\alpha_i} \in \mathcal{U}$ lifts to $\tilde{F}_i: \tilde{U}_{\alpha_i} \rightarrow \tilde{X}^{(p)}$. The single complex $\check{C}(\mathcal{U}, F_*\Omega_{X/S}^\bullet)$ associated to the Čech double complex $\check{C}^q(\mathcal{U}, F_*\Omega_{X/S}^p)$ with differential $\check{d} = d + (-1)^q\delta$, where d is the induced de Rham differential and δ is the alternating difference of restriction to the q -fold intersection $U_{\alpha_0 \dots \alpha_{q-1}} := U_{\alpha_1} \cap \dots \cap U_{\alpha_{q-1}}$ (see [BT13, §8]), has components

$$\check{C}(\mathcal{U}, F_*\Omega_{X/S}^\bullet)^n = \bigoplus_{p+q=n} \check{C}^q(\mathcal{U}, F_*\Omega_{X/S}^p).$$

In particular, the component in degree 1 is given by

$$\check{C}(\mathcal{U}, F_*\Omega_{X/S}^\bullet)^1 = \check{C}^1(\mathcal{U}, F_*\mathcal{O}_X) \oplus \check{C}^0(\mathcal{U}, F_*\Omega_{X/S}^1).$$

Since $\check{C}(\mathcal{U}, F_*\Omega_{X/S}^q)$ is a resolution for $F_*\Omega_{X/S}^q$ for all q , $\check{C}(\mathcal{U}, F_*\Omega_{X/S}^q)$ is a quasi-isomorphic augmentation of $F_*\Omega_{X/S}^q$. We thus wish to provide a map into the degree 1 part of the complex

$$h = h_{\mathcal{U}}: \Omega_{X^{(p)}/S}^1 \rightarrow \check{C}^1(\mathcal{U}, F_*\mathcal{O}_X) \oplus \check{C}^0(\mathcal{U}, F_*\Omega_{X/S}^1)$$

such that $\check{d} \circ h = 0$. We set $h = \sum H_{ij}$ on $\check{C}(\mathcal{U}, F_*\mathcal{O}_X) = \bigoplus_{i,j} \Gamma(U_{ij}, F_*\mathcal{O}_X)$ where $U_{ij} = U_i \cap U_j$, and $h = \sum \psi_i^1$ on $\check{C}^0(\mathcal{U}, F_*\Omega_{X/S}^1) = \bigoplus_i \Gamma(U_i, F_*\Omega_{X/S}^1)$. Let $\check{C}^{p,q}$ denote $\check{C}^q(\mathcal{U}, F_*\Omega_{X/S}^p)$. For $\omega \in \Omega_{X^{(p)}/S}^1$, the element $h(\omega)$ can be represented in the double complex as follows:

$$\begin{array}{ccc} \begin{array}{ccc} \check{C}^{0,2} & & 0 \\ \uparrow -\delta & & \uparrow \\ \bigoplus_{i,j} \Gamma(U_{ij}, F_*\mathcal{O}_X) & \xrightarrow{d} & \check{C}^{1,1} \\ \uparrow \delta & & \uparrow \delta \\ \check{C}^{0,0} & \xrightarrow{d} & \bigoplus_i \Gamma(U_i, F_*\Omega_{X/S}^1) \end{array} & \xrightarrow{d} & \begin{array}{ccc} \sum H_{ij}(\omega) & \mapsto & (dH_{ij} - \psi_i^1 + \psi_j^1)_{i,j} \\ & & \uparrow \\ & & \sum \psi_i^1(\omega) \end{array} \\ & & \xrightarrow{\quad} 0 \end{array}$$

We see that $\sum H_{ij}(\omega)$ maps to 0 from step 2 and $\sum \psi_i^1(\omega)$ also maps to 0 from step 1. It follows that $dH_{ij} - \psi_i^1 + \psi_j^1 = 0$ by step 2 for each pair of indices i, j , thus showing that $\check{d} \circ h = 0$. It can also be checked this is independent of two different covers $\mathcal{U}$ and $\mathcal{V}$ of X by considering the cover $\mathcal{U} \amalg \mathcal{V}$ (see [DI87]).

Step 4. Fix a lifting $\tilde{X}^{(p)}$ of $X^{(p)}$. We define the map $\psi^i: \Omega_{X^{(p)}/S}^i[-i] \rightarrow F_*\Omega_{X/S}^\bullet$ via the compositions

$$\Omega_{X^{(p)}/S}^i[-i] \xrightarrow{s} (\Omega_{X^{(p)}/S}^1)^{\otimes i}[-i] \cong (\Omega_{X^{(p)}/S}^1[-1])^{\otimes i} \xrightarrow{h^{\otimes i}} (F_*\Omega_{X/S}^\bullet)^{\otimes i} \xrightarrow{\text{mult}} F_*\Omega_{X/S}^\bullet.$$

The first map is the antisymmetrisation section

$$s(\omega_1 \wedge \dots \wedge \omega_i) = \frac{1}{i!} \sum_{\sigma \in S_i} \text{sgn}(\sigma) \omega_{\sigma(1)} \wedge \dots \wedge \omega_{\sigma(i)}$$

which is well-defined as $i < p$, the ext map is an isomorphism as $\Omega_{X^{(p)}/S}^1$ is locally free of finite type, and the last is the multiplication map. Since the antisymmetrisation arrow is a section of the projections of $(\Omega_{X^{(p)}/S}^1)^{\otimes i}$ onto $\Omega_{X^{(p)}/S}^i$, the multiplicative property of the Cartier isomorphism results in ψ^i inducing C^{-1} on $\mathcal{H}^i$. $\square$

If $Y = \text{Spec}(k)$ where k is a perfect field of characteristic k , then $\tilde{Y} = \text{Spec}(W_2(k))$ lifts Y over $\mathbb{Z}/p^2\mathbb{Z}$. If X is a smooth Y -scheme, then lifting X over $\tilde{Y}$ is equivalent to lifting $X^{(p)}$ as the absolute Frobenius map of Y is an automorphism. We have thus deduced the following:

Corollary 5.3. *Let k be a perfect field of characteristic p , and let X be a smooth and proper k -scheme such that $\dim(X) < p$. If X is lifted over $W_2(k)$, the Hodge to de Rham spectral sequence*

$$E_1^{p,q} = H^q(X, \Omega_{X/k}^p) \Rightarrow H_{dR}^{p+q}(X/k)$$

degenerates at E_1 .

6. FIELDS OF CHARACTERISTIC ZERO

Having proved the main technical portion of [DI87], we now outline a sketch of how the authors extended their argument to the case where X is a k -scheme of characteristic zero (see [DI87] for full details). Their idea was to write k as an inductive limit of its $\mathbb{Z}$ -subalgebras of finite type, allowing the “spreading out” of certain data, and then using density properties of closed points of schemes to create a lift over $\text{Spec } W_2(k)$ and apply Corollary 5.3.

Remark 6.1. In characteristic zero, Serre’s GAGA principle allows one to use the equivalence of Hodge and de Rham cohomology for smooth complex varieties, such as complex manifolds, to provide a comparison between the analytic and algebraic cohomology theories (see [Gro66b]). However, this does not translate to the case where k is positive characteristic so Deligne and Illusie’s technique circumvents this issue.

Theorem 6.2. *Let k be a field of characteristic zero, and X a smooth and proper k -scheme. Then the Hodge to de Rham spectral sequence degenerates at E_1 .*

Proof sketch. Set $h^{i,j} := \dim_k H^j(X, \Omega_{X/k}^i)$ and $h^n := \dim_k H_{dR}^n(X/k)$. It suffices to show that $h^n = \sum_{i+j=n} h^{i,j}$ by Lemma 3.6. Write k as an inductive limit of the family $k = \varinjlim_{\alpha \in I} A_\alpha$, where the inductive limit is taken over subalgebras of k that are finitely generated over $\mathbb{Z}$. Then there exists some A in this collection and a smooth and proper S -scheme $\mathfrak{X}$, where $S = \text{Spec } A$, such that $X \rightarrow \text{Spec } k$ is obtained via base change. Let $f: \mathfrak{X} \rightarrow S$.

Since S is a noetherian integral scheme of finite type, there exists an affine open subscheme $S' \subset S$ where the coherent sheaves $R^j f_* \Omega_{\mathfrak{X}/S}^i$ and $\mathbb{R}^n f_* \Omega_{\mathfrak{X}/S}^\bullet$, given by $s \mapsto \dim_k H^j(\mathfrak{X}_s, \Omega_{\mathfrak{X}_s}^i)$ and $s \mapsto \dim_k H^n(\mathfrak{X}_s, \Omega_{\mathfrak{X}_s}^\bullet)$ respectively, are locally free on S . Here, R^j and $\mathbb{R}^n$ denote the (right-)derived functors of the pullback, and $\mathfrak{X}_s$ denotes the fibre of $\mathfrak{X}$ over s .

Pick a closed point s such that $s = \text{Spec } k' \rightarrow S$ has large enough residue characteristic p , which lifts to $\tilde{s}: \text{Spec } W_2(k') \rightarrow S$ by the smoothness of S . We thus have the diagram:

$$\begin{array}{ccccccc} \mathfrak{X}_s & \longrightarrow & \mathfrak{X}_{\tilde{s}} & \longrightarrow & \mathfrak{X} & \longleftarrow & X \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \text{Spec } k' & \longrightarrow & \text{Spec } W_2(k') & \longrightarrow & S & \longleftarrow & \text{Spec } k \end{array}$$

By construction, $\mathfrak{X}_s$ is a smooth and proper k' -scheme of dimension $< p$, lifted over $\text{Spec } W_2(k)$. Therefore, the Hodge to de Rham spectral sequence of $\mathfrak{X}_s$ over k' degenerates at E_1 by Corollary 5.3, that is,

$$\dim_{k'} H_{dR}^n(\mathfrak{X}_s, \Omega_{\mathfrak{X}_s/k'}^\bullet) = \sum_{i+j=n} \dim_{k'} H^j(\mathfrak{X}_s, \Omega_{\mathfrak{X}_s/k'}^i).$$

However, this is equivalent to showing

$$\mathrm{rank} \, \mathbb{R}^n f_* \Omega_{\mathfrak{X}/S}^\bullet = \sum_{i+j=n} \mathrm{rank} \, R^j f_* \Omega_{\mathfrak{X}/S}^i,$$

which is the same as $h^n = \sum_{i+j=n} h^{i,j}$. □

Taking Corollary 5.3 and Theorem 6.2 together thus proves Theorem 1.1.

REFERENCES

- [Bei24] Yangda Bei. Hodge decomposition and the degeneration of the frölicher spectral sequence. 2024.
- [BT13] Raoul Bott and Loring W Tu. *Differential forms in algebraic topology*, volume 82. Springer Science & Business Media, 2013.
- [DI87] Pierre Deligne and Luc Illusie. Relèvements modulo p et décomposition du complexe de de rham. *Inventiones mathematicae*, 89:247–270, 1987.
- [Gro61] Alexander Grothendieck. Éléments de géométrie algébrique: Iii. étude cohomologique des faisceaux cohérents, première partie. *Publications Mathématiques de l’IHÉS*, 11:5–167, 1961.
- [Gro66a] Alexander Grothendieck. Éléments de géométrie algébrique: Iv. étude locale des schémas et des morphismes de schémas, troisième partie. *Publications Mathématiques de l’IHÉS*, 28:5–255, 1966.
- [Gro66b] Alexander Grothendieck. On the de rham cohomology of algebraic varieties. *Publications Mathématiques de l’Institut des Hautes Études Scientifiques*, 29(1):95–103, 1966.
- [Har13] Robin Hartshorne. *Algebraic geometry*, volume 52. Springer Science & Business Media, 2013.
- [Ill02] Luc Illusie. Frobenius and hodge degeneration. *Introduction to Hodge theory*, 8:99–149, 2002.
- [Kat70] Nicholas M Katz. Nilpotent connections and the monodromy theorem: Applications of a result of turrittin. *Publications mathématiques de l’IHÉS*, 39:175–232, 1970.
- [Ser56] Jean-Pierre Serre. Géométrie algébrique et géométrie analytique. In *Annales de l’institut Fourier*, volume 6, pages 1–42, 1956.
- [Voi02] Claire Voisin. *Hodge Theory and Complex Algebraic Geometry I: Volume 1*, volume 76. Cambridge University Press, 2002.